

Lab 4 – Vibe Coding

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Section: G

Game title: Air Hockey

Repo link:-

Forked game repo: https://github.com/rekhadhorigol/02_air_hockey.git

Lab-4 repo: <https://github.com/rekhadhorigol/SE-Labs-PES1UG24CS370.git>

Things Done:-

1) Deliberate Bug fixed:

The original puck-versus-paddle collision implementation did not correctly handle collision direction. It could simply reverse a velocity component, which could cause incorrect behaviour for angled collisions and could allow the puck to remain overlapped with a paddle.

- Calculated the collision normal between the puck & paddle.
- Pushed the puck outside the paddle after an overlap.
- Reflected puck velocity using the collision normal.
- Reflected velocity only when the puck was travelling toward the paddle.
- Used smaller puck movement substeps to reduce tunnelling through paddles at higher speeds.

Files updated: game/collisions.py and game/puck.py.

2) Match Scoring:

Implemented scoring for both sides of the Air Hockey table. When the puck enters the left goal, the computer score is incremented. When the puck enters the right goal, the player score is incremented. The current scores are displayed during gameplay.

- player_score & computer_score are initialized at the start of a match.
- Goal detection increments the appropriate score.
- Player & computer scores are displayed on screen.
- Goal detection is restricted to the actual goal opening, the wall outside the opening still bounces the puck.

3) 30-second match timer:

Implemented a 30-second match timer. The remaining time is calculated from the match start time and displayed during gameplay. When the timer reaches zero, the match ends and the final result is displayed.

- Match duration is set to 30 seconds.
- Remaining time is displayed at the top of the game.
- Gameplay stops when the timer reaches zero.
- The final result is determined from the player and computer scores.

4) Puck reset after goal:

Implemented the required puck reset behaviour after every goal. The puck is returned to the centre of the table and immediately receives a new velocity, so it continues play without remaining motionless.

- Puck position is reset to the centre of the table after a goal.
- A new direction and velocity are assigned immediately.
- The puck resumes play automatically after scoring.

Testing & verification:-

The completed application was tested after implementation for the following behaviours:

Test	Expected Result	Status
Paddle collision	Puck rebounds correctly, including angled collisions.	PASS
Left goal	Computer score increases by 1.	PASS
Right goal	Player score increases by 1.	PASS
Non-goal end wall	Puck bounces without changing the score.	PASS
Puck reset	Puck returns to centre & immediately continues moving.	PASS
30-second timer	Timer reaches zero & the match ends.	PASS
Final result	Win/loss/draw is displayed based on the score.	PASS

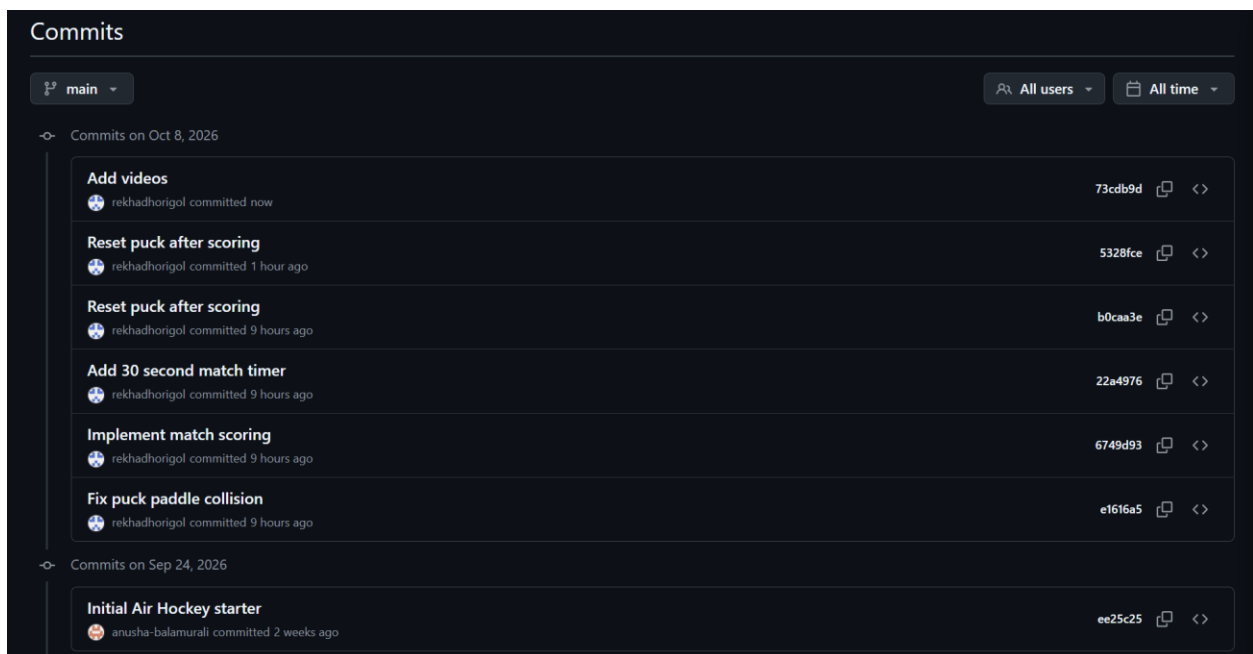
Before & After videos:-

A 10-second video was recorded before making changes to demonstrate the original application. A second 10-second video was recorded after all fixes & features were implemented.

Git Commit evidence:-

The implementation was maintained with separate commits for the individual tasks as required by the lab handout.

Task	Commit Message
Task 1 – Deliberate bug fix	Fix puck paddle collision
Task 2 – Scoring	Implement match scoring
Task 3 – Timer	Add 30 second match timer
Task 4 – Puck reset	Reset puck after scoring



(By mistake committed task 4 twice, actually the last commit, commit 5 of my git username, was for final check I had done)

Prompt (Chat history) evidence:

Prompt evidence was maintained through the LLM chat(ChatGPT) used during the implementation. The chat contains the task requirements, project analysis, implementation guidance, debugging of the goal-detection issue & final verification.

Chat history / prompt evidence link: <https://chatgpt.com/share/6ac7a0a1-1aec-83ee-bedc-705605d0958b>

Final things done checklist:-

- Task 1 - Deliberate collision bug fixed.
- Task 2 - Match scoring implemented.
- Task 3 - 30-second timer implemented.
- Task 4 - Puck reset after scoring implemented.
- Separate Git commits maintained for each task.
- 10-second before video recorded.
- 10-second after video recorded.
- PDF document prepared from this Word document for submission.
- Chat/prompt evidence retained.

The Air Hockey game was successfully updated using LLM-assisted vibe coding. The deliberate puck-paddle collision bug was fixed & the required scoring, 30-second timer, and post-goal puck reset features were implemented & tested. The final application was verified to behave correctly and the updated code was pushed to the individual forked repository with separate commits for the required tasks.